

CE 310

Correlation — When Variables Move Together

Week 7 · Session 1 · The Building as a System of Relationships

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Part 1

A Confounding Mistake

The Daylighting Trap

90 seconds · what went wrong here?

Phoenix, 2017. A building owner spends **\$85,000** on a full-glass south facade, specifically to cut lighting energy by letting daylight do the work.

A year of bills comes back. Lighting energy is **highest in summer, lowest in winter** — and the correlation between month and lighting energy is strong.

The owner's consultant reads the data and reports:

"The daylighting strategy works in winter and is ineffective in summer. Recommend adding shading to recover summer performance."

The correlation is real and the arithmetic is correct. The conclusion is still wrong.

Why? Talk to a neighbor for 90 seconds before the next slide.



Nothing Connected the Window to the Light Switch

The building had no daylight sensors and no dimming controls.

Glass alone cannot reduce lighting energy. The fixtures run at full output whenever the space is occupied — bright noon or overcast dusk, the wattage is identical. Without a sensor, the facade had no mechanism to act through at all.

So what did the bills actually track? **Operating hours.** This building ran an extended summer schedule, so summer occupancy hours were longest — and lighting energy followed the schedule, not the sun.



The consultant's error was not statistical. Their r was computed correctly. They assumed the variable they cared about (daylight) was the one driving the pattern, when a third variable (operating hours) was driving it and daylight was driving nothing.



Correlation describes a pattern between two measured variables. It cannot tell you which one causes which, or whether a hidden third variable — a **confounder** — is driving both. Always ask: **what is the physical mechanism?** If you cannot name one, you do not have a cause.

Week 6 Looking Ahead — Answer

Moving together, or just sharing a spreadsheet?

Week 6 asked: **How do you tell whether two variables actually move together, versus just happening to sit in the same spreadsheet?**

Answer: The **Pearson correlation coefficient** r — one number, ranging -1 to $+1$.

Same file, same 8,760 hours, same building. Outdoor temperature vs. cooling energy: the Pearson $r = 0.7752$ — hotter hours, higher cooling loads.

Zone temperature vs. cooling energy: $r = 0.2764$. Sharing a spreadsheet is not a relationship.

Today's Tools — Five Concepts

By the end of this lecture you will be able to:

1. **Pearson r** — measure strength and direction of linear co-movement
2. r^2 — fraction of one variable's variance explained by the other
3. **Covariance** — the raw joint variation before standardizing (`COVARIANCE.S` function)
4. **Conditional correlation** — r computed on a subgroup (season, operating mode)
5. **Scatter plot** — visual tool for seeing the shape of the relationship

This week's dataset: `Week5_D0E_Med0ffice_ZoneTemp_2023.csv` — download from D2L (8,760 hourly rows)

- Column G = `Zone_Temp_F` · Column H = `Outdoor_Temp_F` · Column I = `Cooling_kWh`

Part 2

Measuring Co-Movement

What Correlation Captures

What r measures:

- Direction of the linear relationship (positive or negative)
- How consistently two variables move together
- Dimensionless — no units
- Ranges from -1 (perfect negative) to $+1$ (perfect positive)
- $r = 0$ means no linear relationship

What r does NOT tell you:

- Which variable causes which
- Whether the relationship is non-linear (a curved relationship can have $r \approx 0$)
- Whether a third variable is driving both
- What happens on any individual hour

A strong r does not mean causation. It means two variables tend to move together in the data.

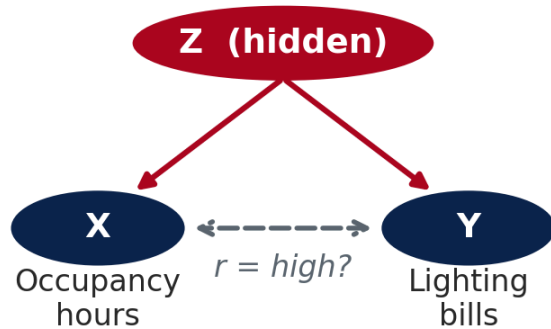
Correlation $\neq$ Causation — Three Structures

1. Confounder — a hidden Z drives both X and Y . Remove Z and the apparent X - Y correlation weakens or disappears. (*Daylighting trap: occupancy hours drove both lighting bills and apparent daylighting performance.*)

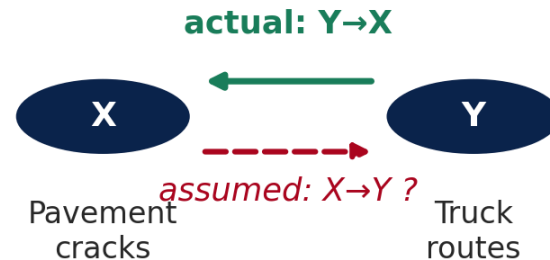
2. Reverse causation — $Y \rightarrow X$, not $X \rightarrow Y$. The correlation is real, but the assumed direction is backward. (*Heavy trucks correlate with pavement damage — but trucks cause the cracks, not the reverse.*)

Three Structural Reasons $r \neq$ Causation

1. Confounder
 Z drives both — no $X \rightarrow Y$ link

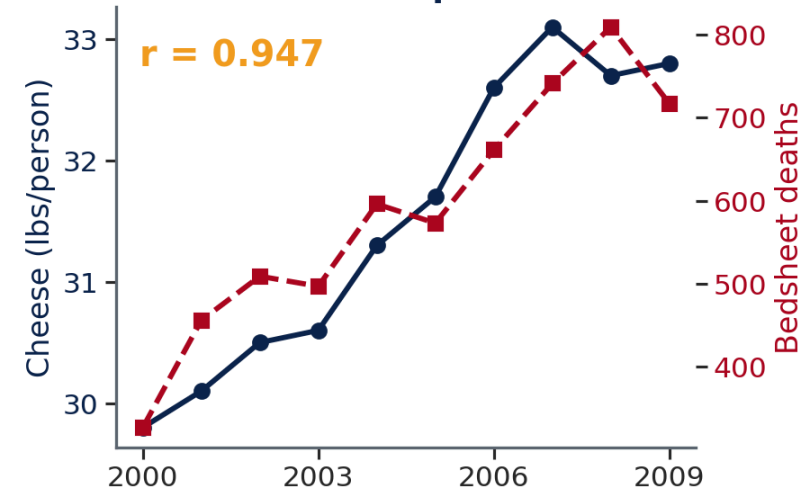


2. Reverse Causation
 $Y \rightarrow X$, not $X \rightarrow Y$



"Cracks attract trucks?" No — trucks create cracks.

3. Coincidence
Two unrelated upward trends



Correlation $\neq$ Causation — Three Structures (cont.)

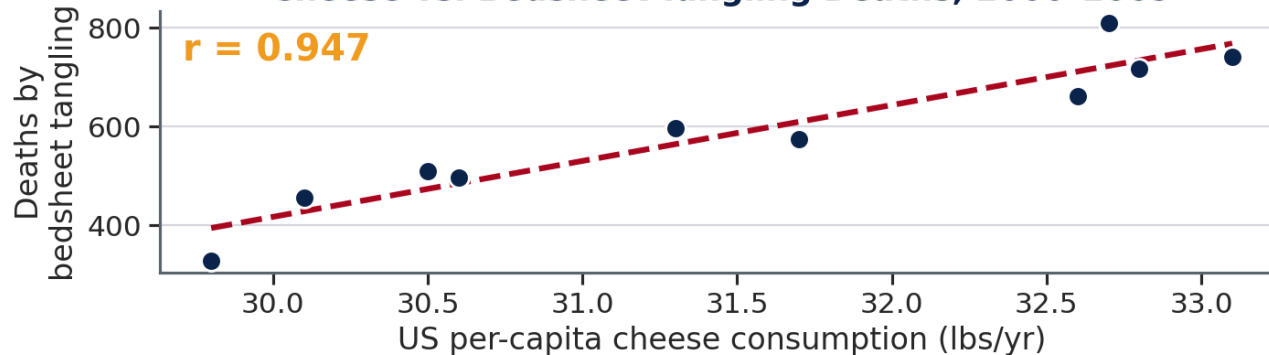
3. Coincidence — two unrelated variables both trend over the same period. $r = 0.95$, mechanism = none.

High r cannot tell you which of these three is true. Only domain knowledge — the physical mechanism — resolves it.

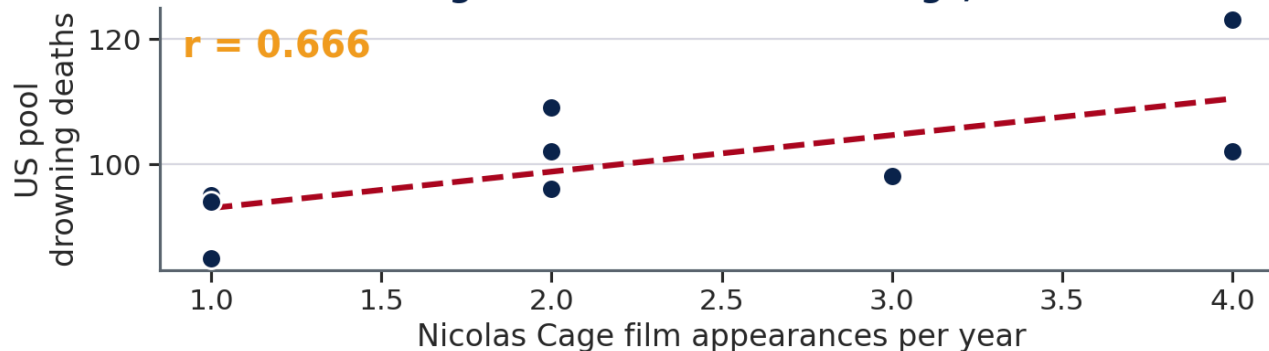
Famous Spurious Correlations

High r — Zero Mechanism (Tyler Vigen, 2015; USDA; CDC)

Cheese vs. Bedsheet-Tangling Deaths, 2000-2009



Nicolas Cage Films vs. Pool Drownings, 1999-2009



Both pairs have **strong Pearson r** — the cheese–bedsheet pair ($r = 0.947$) exceeds our own outdoor-cooling pair ($r = 0.775$).

Neither has a physical mechanism.

Why? Any two series that both trend upward over the same years produce a high r — the algorithm cannot tell a physical link from a coincidence of trends.

💡 **The question r cannot answer:** "Is there a mechanism by which X physically produces Y ?" — engineering judgment, not Excel.

The Pearson r Formula

$$r = \frac{\text{cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

Numerator: covariance — how X and Y vary together (has units)

Denominator: product of standard deviations — removes units, scales result to $[-1, +1]$

Covariance — what the numerator measures:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

When X and Y are both above their means simultaneously $\rightarrow$ positive product $\rightarrow$ positive contribution. Summed over all n observations, the sign and size capture the dominant direction of co-movement. This is the variance formula, generalized to two variables.

The Pearson r Formula (cont.)

Applied to outdoor temperature and cooling energy:

$$r = \frac{331.58 \text{ }^{\circ}\text{F} \cdot \text{kWh}}{16.918 \text{ }^{\circ}\text{F} \times 25.282 \text{ kWh}} = \frac{331.58}{427.6} = 0.7752$$

Why the units cancel: the numerator carries $^{\circ}\text{F} \cdot \text{kWh}$; the denominator carries $^{\circ}\text{F} \times \text{kWh}$. They divide out, leaving r a pure number with no units at all.

That is what bounds r to $[-1, +1]$ — and what makes it comparable across domains. Measure cooling in BTU instead of kWh and r is unchanged, even though the raw covariance changes by a factor of 3,412.

How to Read r

Abs(r)	Strength	Engineering meaning
0.9 – 1.0	Very strong	Nearly perfect linear predictor
0.7 – 0.9	Strong	Reliable for preliminary models
0.5 – 0.7	Moderate	Useful but significant unexplained variation
0.3 – 0.5	Weak	Poor linear predictor
0.0 – 0.3	Negligible	No linear relationship

Sign: Positive $r \rightarrow Y$ rises with X ; Negative $r \rightarrow Y$ falls with X .

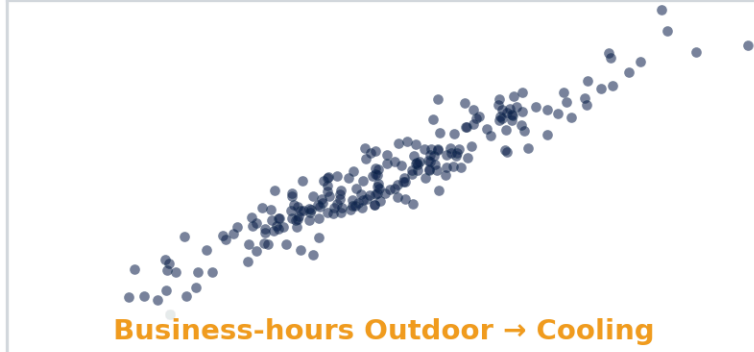
$r = 0.7752$: **strong positive** — outdoor temperature is a reliable predictor of cooling load.

What Those Bands Actually Look Like

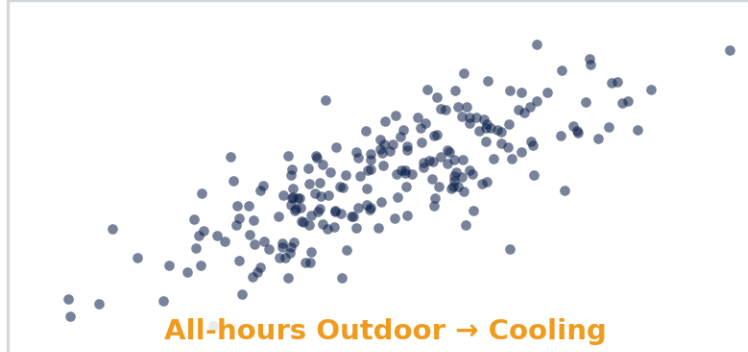
Same five bands as the table — now as clouds of points. Three of these panels are pairs from this week's own dataset.

The Same Five Bands, Drawn

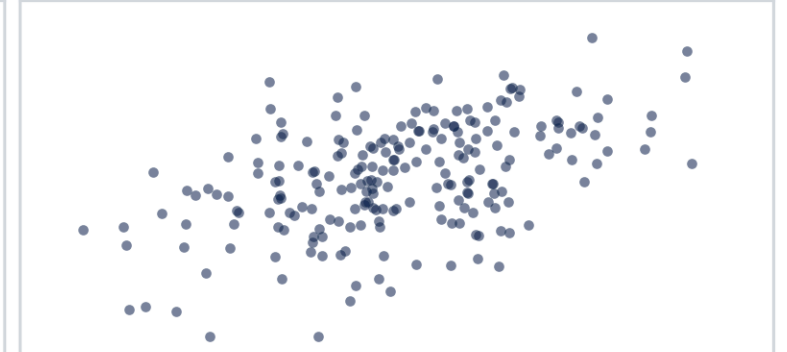
$r = +0.95$ Very strong



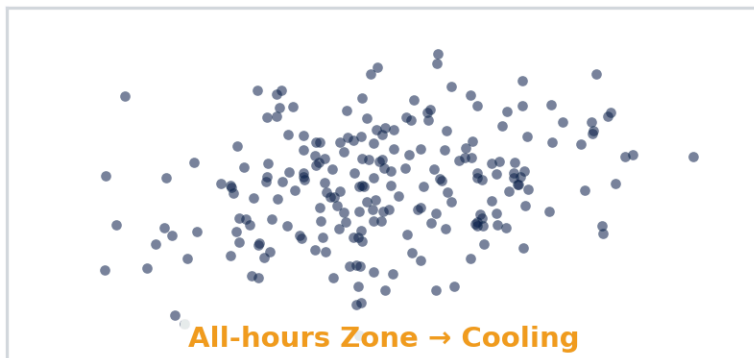
$r = +0.78$ Strong



$r = +0.50$ Moderate



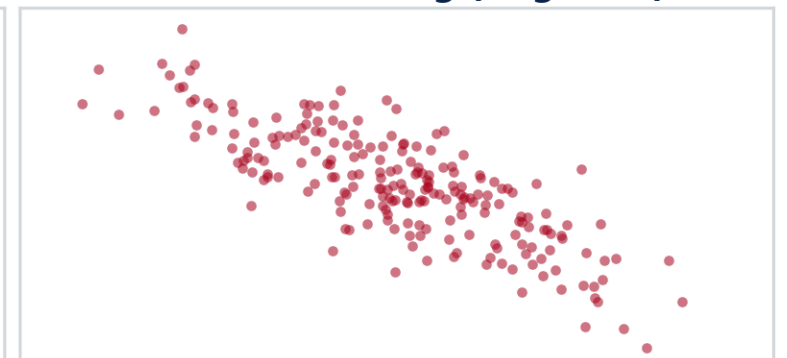
$r = +0.28$ Weak



$r = +0.00$ Negligible



$r = -0.80$ Strong (negative)



Part 3

Three Pairs, One Surprise

Correlation Across CE Domains

Pearson r is a universal tool — the same formula applies across every CE domain:

Domain	Variable Pair	Typical r	Direction	Physical reason
Materials / Structural	Water-cement ratio → 28-day compressive strength	≈ -0.90	Negative	More water dilutes cement paste → weaker concrete (ACI 318)
Transportation	Freeway vehicle density → average speed	≈ -0.75	Negative	More vehicles → congestion → slower flow (HCM Level of Service)
Hydrology	Rainfall intensity → runoff coefficient	$\approx +0.82$	Positive	Faster rain exceeds soil infiltration rate → more runoff (NRCS TR-55)
Building Energy	Outdoor temp → cooling load	0.78	Positive	Hotter weather → more HVAC work (this week's dataset)

Correlation Across CE Domains (cont.)

The sign and magnitude of r encode the physical mechanism. A plausible mechanism is also what separates a useful engineering model from a spurious coincidence.

Predict — Which Pair Is Strongest?

45 seconds · rank all three

Same building, same 8,760 hours. Three pairs of columns:

- **Outdoor_Temp_F vs. Cooling_kWh**
- **Zone_Temp_F vs. Outdoor_Temp_F**
- **Zone_Temp_F vs. Cooling_kWh**

Rank them by correlation strength, strongest first. Commit to an order before the next slide.

Three Correlation Pairs

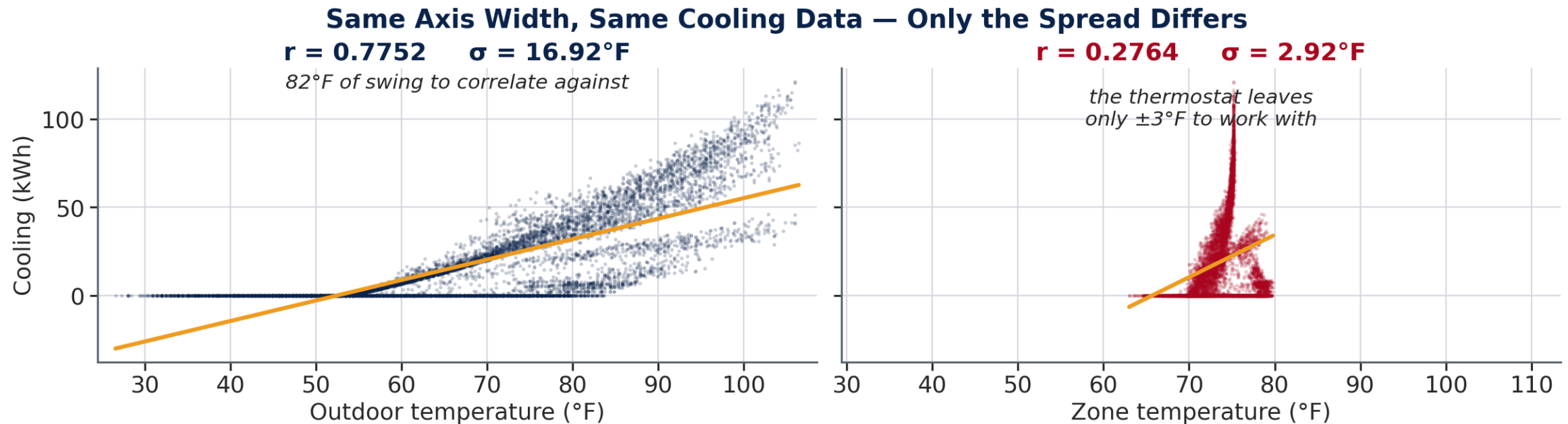
Pair	r	Interpretation
Outdoor_Temp_F vs. Cooling_kWh	0.7752	Strong positive — main pair for regression
Zone_Temp_F vs. Outdoor_Temp_F	0.7529	Moderate-strong — HVAC limits zone temperature swing
Zone_Temp_F vs. Cooling_kWh	0.2764	Weak — zone temp is a poor predictor of cooling load

Largest r : Outdoor vs. Cooling (0.7752). **Weakest:** Zone vs. Cooling (0.28) — not the pair most students expect.

Students often expect zone-cooling to be the strongest pair: "if the zone is hot, HVAC must be working hard." The data says otherwise. The next slide explains why.

Why Zone-Cooling r is Weak – HVAC Control

Intuition: "Hot zone → high cooling?" Direction yes ($r = +0.28$) — but zone temp is the **result** of HVAC winning. Both panels below span the same 84°F of axis.



Tight control makes zone temp a **thermostat success metric** — not a predictor of HVAC effort.

r^2 — The Coefficient of Determination

r^2 is the fraction of variance in Y explained by X :

$$r^2 = \frac{[\text{cov}(X, Y)]^2}{\text{var}(X) \cdot \text{var}(Y)}$$

Pair	r	r^2	Interpretation
Outdoor vs. Cooling	0.7752	0.6010	Outdoor temp explains 60.1% of cooling variance
Zone vs. Cooling	0.2764	0.0764	Zone temp explains only 7.6% of cooling variance

Outdoor temp is roughly **8x more informative** than zone temp (60.1% vs. 7.6%).

The remaining 39.9% unexplained by outdoor temp alone includes:

- Solar gain variation (cloud cover, orientation)

Which Predictor Would You Keep?

Computing r between every pair of candidate predictors is the correlation-matrix step of the **exploratory data analysis** routine from Week 2 — the check a modeler runs before fitting anything.

Rule: if two features have $|r| > 0.9$, they carry nearly identical information — including both creates **multicollinearity**: inflated standard errors, unstable coefficient estimates, unreliable predictions.

Feature pair	Abs(r)	Multicollinearity risk
Outdoor_Temp_F vs. Month	0.14	None — essentially independent as predictors
Zone_Temp_F vs. Outdoor_Temp_F	0.75	Moderate — some overlap but tolerable
Outdoor_Temp_F vs. Cooling_kWh	0.78	Not applicable — predictor vs. response

Week 9 uses outdoor temperature alone — not because Month creates multicollinearity, but because it adds almost no predictive power (0.9% vs. outdoor's 60.1%).

Quick Check — Correlation and Causation

Think, then check with your neighbor

Question A: If $r(\text{Outdoor, Zone}) = 0.75$ and $r(\text{Outdoor, Cooling}) = 0.78$, does that mean outdoor temperature is a *cause* of both zone temperature and cooling energy?

Question B: A building in Las Vegas has $r(\text{Outdoor, Cooling}) = 0.92$; for a building in Seattle, $r = 0.61$. Which climate has a more consistent outdoor-cooling relationship, and why?

Compare with your neighbor.

✓ Correlation and Causation — Answers

Question A: Correlation does NOT imply causation — but outdoor temperature DOES physically drive cooling through thermodynamics. The r value tells you only about statistical co-movement. The physical mechanism (conduction, convection, solar gain) provides the causal link, but r alone cannot establish it. Both statements can be true simultaneously: high r AND a real causal mechanism.

Question B: Las Vegas has a more consistent outdoor-cooling relationship because it operates in a single dominant regime — hot, dry, high solar gain year-round. Seattle's moderate climate has mixed heating/cooling seasons, maritime cloud variability, and mild shoulder seasons that add scatter to the outdoor-cooling relationship.

Climate	$r(\text{Outdoor, Cooling})$	Why
Las Vegas (hypothetical)	0.92	Single dominant cooling regime
Tucson (actual)	0.78	Mixed: cooling season + ~3,700 chiller-off hours

Part 4

When r Changes by Subgroup

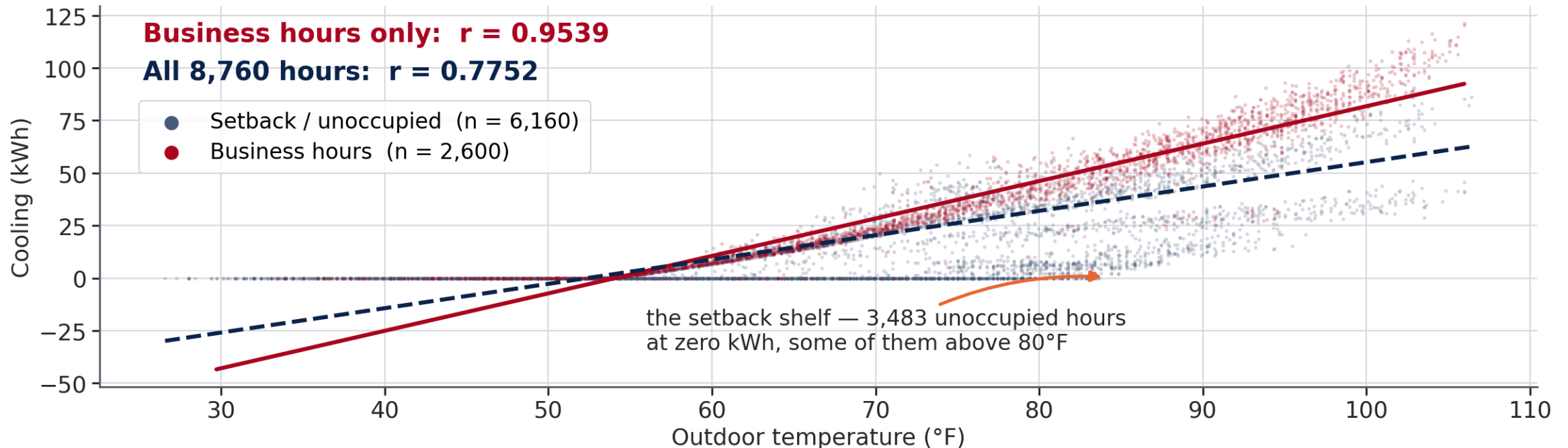
Why Condition?

The all-hours $r(\text{Outdoor, Cooling}) = 0.7752$ mixes 8,760 hours with very different operating modes: occupied vs. unoccupied, summer vs. winter, business hours vs. setback.

Subgroup	$r(\text{Outdoor, Cooling})$	n	What this means
All hours	0.7752	8,760	Blended average of all modes
Business hours	0.9539	2,600	Nearly perfect — active control mode
Non-business hours	0.6864	6,160	Weaker — setback reduces correlation
Spring	0.8143	2,208	Cleanest season (no monsoon scatter)
Summer	0.7494	2,208	Lowest seasonal r (monsoon adds scatter)

Business Hours — r Jumps to 0.95

Filtering to occupied hours gives $r^2 = 0.9100$ — outdoor temperature explains **91% of cooling variance**.



The all-hours $r = 0.78$ answers "what is r for the full year?" — not "what is r during active HVAC mode?" Choose based on your design question.

Seasonal r — Spring Highest, Summer Lowest

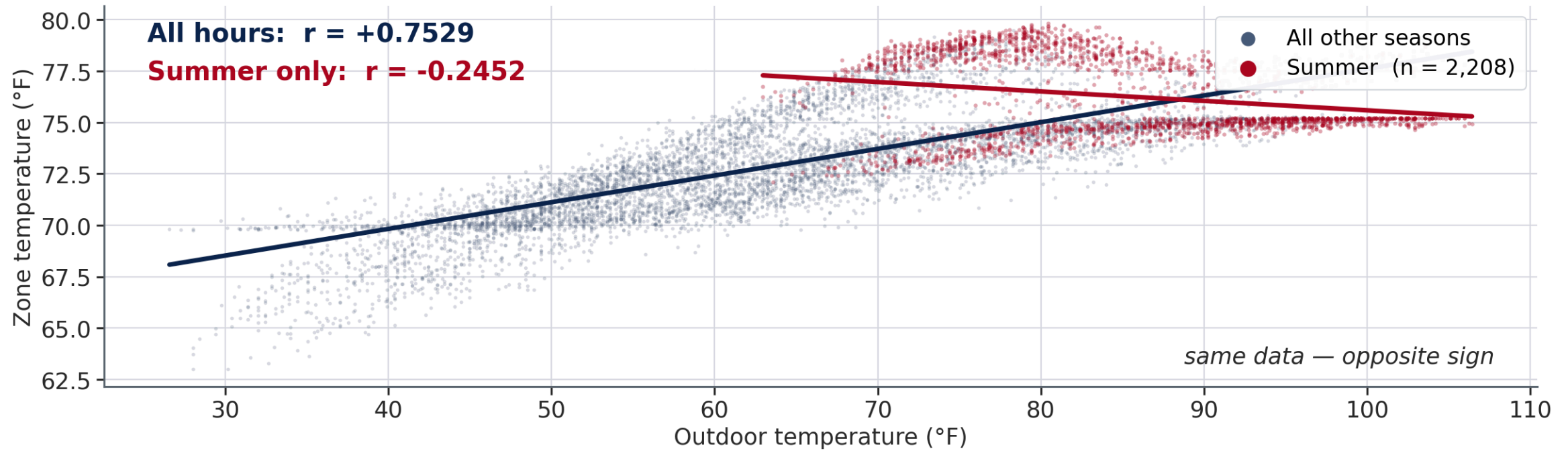
Season	r	n	Why
Spring	0.8143	2,208	Clean cool-to-warm sweep — near-monotone, strong linear signal
Fall	0.7524	2,184	Symmetric to spring — similarly clean sweep
Winter	0.7541	2,160	Real cooling still occurs from internal gains
Summer	0.7494	2,208	LOWEST — monsoon humidity and solar gain add scatter

Range: $\text{MAX} - \text{MIN} = 0.8143 - 0.7494 = 0.0649$

Only 0.065 range across all four seasons — the outdoor-cooling r is essentially the same year-round.
One regression model (Week 9) works for all seasons without seasonal adjustment.

Simpson's Paradox – The Negative Summer r

$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ for **summer only** = -0.2452



Simpson's Paradox: the direction of a relationship can **reverse** when you condition on a subgroup. At 108°F the chiller runs flat out and drives zone temp *down*.

Month is Not a Physical Predictor

$$r(\text{Month}, \text{Outdoor_Temp_F}) = 0.1438 \cdot r(\text{Month}, \text{Cooling_kWh}) = 0.0942$$

Both near zero — yet we know temperature is strongly seasonal. **Why?**

Pearson r only detects **monotone linear** relationships. Outdoor temperature in Tucson rises January → July then falls July → December. This parabolic (up-then-down) shape averages out to near-zero linear correlation with month (1–12).

Predictor	r with Cooling_kWh	r^2	Use in regression?
Month (1–12)	0.0942	0.009	No — non-linear pattern invisible to Pearson r
Outdoor_Temp_F	0.7752	0.601	Yes — physical mechanism, strong linear signal

Always use the **physical mechanism variable**, not a calendar label. Month encodes seasonality indirectly and non-linearly; outdoor temperature encodes the same information directly and linearly.

Quick Check B — Simpson's Paradox and Non-Linearity

Think, then check with your neighbor

Question A: $r(\text{Zone}, \text{Outdoor})$ for all hours is $+0.75$, but for summer it is -0.25 . Can both be right at the same time?

Question B: If a colleague says "the correlation between month and cooling is only 0.09, so there's no seasonality," how would you respond?

Compare with your neighbor.



Quick Check B — Simpson's Paradox

Both r values are correct — they answer different questions

Question A: Yes — both can be correct at once. The all-hours $r = +0.75$ is driven by the seasonal mean difference: winter zone temps cluster at 68°F, summer at 75°F, so the two trend together across the year. Within summer, the HVAC control reversal dominates — when outdoor temp spikes, the chiller runs harder and zone temp stays flat or drops below setpoint.

Question B: Month as a predictor fails because its relationship with cooling is **parabolic (U-shaped rotated 90°)**: cooling peaks in July, bottoms in December/January. Pearson r only detects monotone linear trends — it cannot find a pattern that goes up then comes back down. The correct tools are AVERAGEIF by month, or a scatter plot of monthly average cooling vs. month number.



When $r \approx 0$, ask: "Could the relationship be non-linear or non-monotone?"

Part 5

From Covariance to Correlation

Covariance — The Raw Signal

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

- When X and Y are **both above** their means: positive product → positive contribution
- When X above and Y below their means: negative product → negative contribution
- Sum of all products → positive cov means they tend to move together

Applied to outdoor temperature vs. cooling:

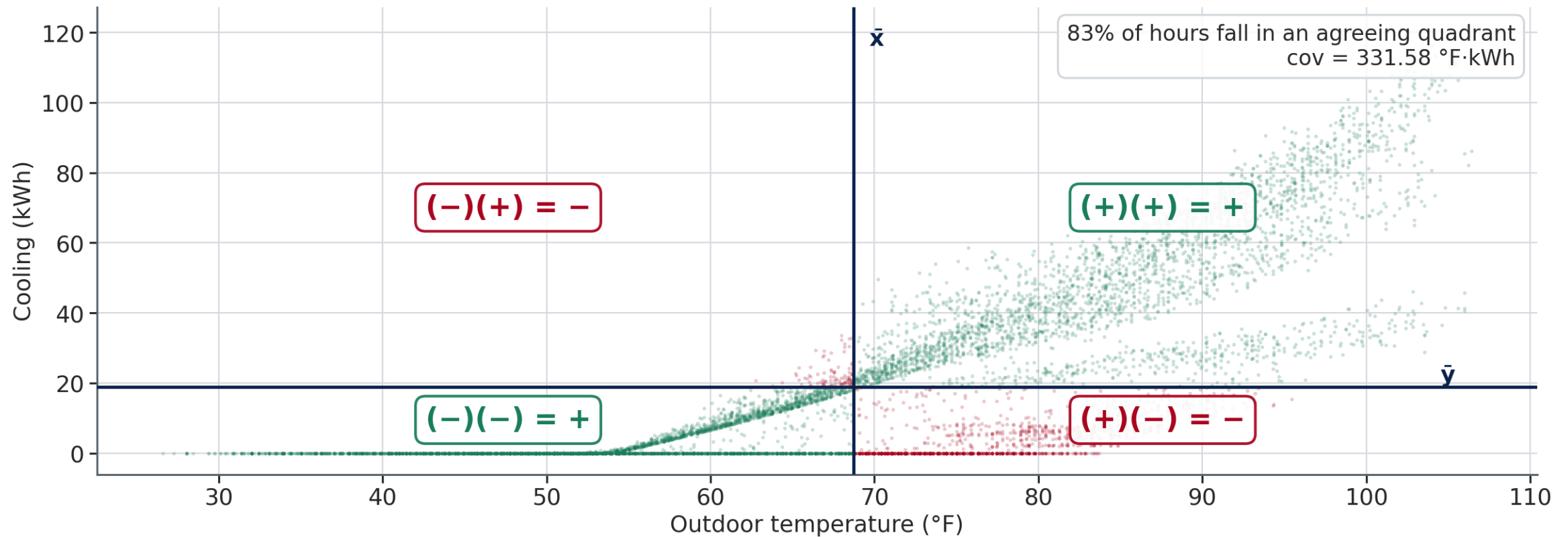
$$\text{cov}(\text{Outdoor}, \text{Cooling}) = 331.58 \text{ } ^\circ\text{F}\cdot\text{kWh}$$

Units: $^\circ\text{F}\cdot\text{kWh}$ — the product of both variables' units. Correlation divides out the units:

$$r = \frac{331.58}{16.918 \times 25.282} = \frac{331.58}{427.6} = 0.7752\checkmark$$

Where the Covariance Sum Gets Its Sign

Split the real cloud at $\bar{x}$ and $\bar{y}$. Each point contributes the product of its two deviations — so the quadrant it lands in decides the sign it adds.



Part 6

Bridge to Machine Learning

Bridge to Machine Learning — Correlation

Three ML algorithms where correlation is the core mathematical operation:

Linear Regression

(Week 9)

$$\beta_1 = r \cdot \frac{\sigma_Y}{\sigma_X}$$

The regression slope **is** correlation, rescaled to the variables' units.

CE: outdoor temp → cooling load model

Principal Component Analysis (PCA)

When predictors are correlated (temperature, humidity, solar gain...), PCA rotates the **correlation matrix** into independent axes — the first captures maximum shared variance.

CE: energy benchmarking — reduce 30 correlated metrics to 3 independent axes

Collaborative Filtering

"Customers who bought X also bought Y." User similarity = Pearson r of rating vectors. Users whose ratings co-move ($r \approx 1$) likely agree on items neither has rated yet.

CE analogy: recommending standard spec sections based on what similar project types used

Three Deliverables This Week

	What it covers	When
Weekly Quiz	15 questions on this lecture and the Tutorial	Closes 8:00 AM Friday, Oct 9
In-Class Exercise	Sections A and B, the charts, and the memo	Work Wed + Fri · due Wed 9:00 AM next week
Homework	A separate take-home problem set	On your own · due Wed 9:00 AM next week

The exercise spans two sessions. You start it Wednesday after the tutorial, and finish it Friday. **You are not expected to finish in class** — use the session time for the parts where having me in the room actually helps, and complete the rest on your own.

💡 Three submissions, three D2L Assignments folders. The quiz closes 8:00 AM Friday, Oct 9; the exercise and the homework are both due Wednesday 9:00 AM — no late work is accepted.

Looking Ahead — Week 8: Midterm Review

Week 8 is the **midterm**. No new dataset, no new Excel function.

Conceptual questions:

- Interpreting r , r^2 , compression ratio
- HVAC control mode effects on correlation
- Simpson's Paradox, confounder identification

All Weeks 1–7 Excel functions are fair game.

See Tutorial Quick Reference slides for the complete list.

Formula writing (apply to dataset):

- NORM.DIST patterns (CDF vs. PDF, upper-tail)
- At-least- k Poisson formulas
- CORREL with subgroup FILTER
- COVARIANCE.S syntax
- Conditional probability from COUNTIFS

Key Results — All-Hours Correlations

Quantity	Value
$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ — all hours	0.7529
$r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ — all hours	0.7752
$r(\text{Zone_Temp_F}, \text{Cooling_kWh})$ — all hours	0.2764
$r^2(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$	0.6010

Key Results — Conditional Correlations

Quantity	Value
$r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ — business hours	0.9539
$r(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$ — non-business hours	0.6864
$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ — Summer only	−0.2452
$r(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$ — Winter only	0.7837
$\text{cov}(\text{Outdoor_Temp_F}, \text{Cooling_kWh})$	331.58 °F·kWh
$\text{cov}(\text{Zone_Temp_F}, \text{Outdoor_Temp_F})$	37.15 °F ²

Week 7 In-Class Exercise — Correlation

Wk07.03_In_Class_Exercise.pdf · Wk07.04_In_Class_Answer_Sheet.xlsx

Section A — Pearson r and r^2 :

- A1–A3: CORREL for all three variable pairs (outdoor-cooling, zone-outdoor, zone-cooling)
- A4: Multiple choice — which pair has the largest $|r|$
- A5: r^2 for the main pair (outdoor vs. cooling)
- A6: Conditional CORREL for business hours using FILTER
- A7–A8: Multiple choice — interpreting r and r^2 values

Section B — Covariance and Conditional Correlation:

- B1–B5: Seasonal r for each season using FILTER